

Impact of nonlinear and polarization effects in coherent systems

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Abstract: Coherent detection with digital signal processing (DSP) significantly changes the ways impairments are managed in optical communication systems. In this paper, we review the recent advances in understanding the impact of fiber nonlinearities, polarization-mode dispersion (PMD), and polarization-dependent loss (PDL) in coherent optical communication systems. We first discuss nonlinear transmission performance of three coherent optical communication systems, homogeneous polarization-division-multiplexed (PDM) quadrature-phase-shift-keying (QPSK), hybrid PDM-QPSK and on/off keying (OOK), and PDM 16-ary quadrature-amplitude modulation (QAM) systems. We show that while the dominant nonlinear effects in coherent optical communication systems without optical dispersion compensators (ODCs) are intra-channel nonlinearities, the dominant nonlinear effects in dispersion-managed (DM) systems with inline dispersion compensation fiber (DCF) are different when different modulation formats are used. In DM coherent optical communication systems using modulation formats of constant amplitude, the dominant nonlinear effect is nonlinear polarization scattering induced by cross-polarization modulation (XPolM), whereas when modulation formats of non-constant amplitude are used, the impact of inter-channel cross-phase modulation (XPM) is much larger than XPolM. We then describe the effects of PMD and PDL in coherent systems. We show that although in principle PMD can be completely compensated in a coherent optical receiver, a real coherent receiver has limited tolerance to PMD due to hardware limitations. Two PDL models used to evaluate PDL impairments are discussed. We find that a simple lumped model significantly over-estimates PDL impairments and show that a distributed model has to be used in order to accurately evaluate PDL impairments. Finally, we apply system outage considerations to coherent systems, taking into account the statistics of polarization effects in fiber.

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1. Introduction

Optical communication systems have experienced tremendous progress over the past few years with the revival of coherent detection technology. Coherent detection received lots of attention in the 1980s because of its high receiver sensitivity [1], which was crucial for regenerated systems prior to the advent of erbium-doped-fiber amplifiers (EDFAs). However, progress on optical amplification technologies together with difficulties in real time implementations of coherent optical receivers caused coherent detection in optical communications to be abandoned in the early 1990s. Using direct-detection technology, 25.6-Tb/s capacity with a spectral efficiency of 3.2 b/s/Hz was achieved in a single fiber, but further increasing the capacity and spectral efficiency using direct-detection became very challenging [2,3]. Advances in digital signal processing (DSP) technology have provided new opportunities for coherent detection in optical communication systems in recent years, and digital coherent optical communications have become the main technology for high-speed and high-capacity optical transport networks [4,5].

As the full optical field information is preserved and accessible after coherent detection, optical phase and polarization can be used to encode data, which significantly increases the spectral efficiency. In digital coherent receivers, signals are converted to the digital format through analog-to-digital-convertors (ADCs) after photo-detection. Therefore carrier phase recovery and polarization alignment and separation, the main obstacles for analog implementations of coherent receivers in earlier years, can be easily realized using sophisticated DSP. Moreover, many impairment compensation functions that are performed in the optical domain in direct-detection systems can be realized in the electrical domain using DSP in coherent detection systems, which significantly changes the ways to manage impairments in optical communication systems. For example, dispersion management, which has been successfully used in direct-detection optical communication systems to reduce fiber nonlinear impairments, is found to be suboptimal in coherent systems [6–10], and polarization-mode dispersion (PMD), which has long been considered as one of the limiting factors for optical communication systems, can be well compensated in digital coherent receivers and is even helpful to reduce fiber nonlinearities [11–19].

With linear impairments being fairly well handled in coherent receives, fiber nonlinearities remain one of the limiting factors of coherent optical communication systems, even if they

have been shown to be partially compensated by DSP in some research demonstrations [20–22]. In direct-detection on-off-keyed (OOK) systems, fiber nonlinearities manifest themselves as timing and amplitude jitter induced by intra-channel and inter-channel nonlinearities [23,24]. In coherent systems, as information is also coded in phase and polarization, depending on system parameters such as bit rate and modulation format, phase and polarization distortions caused by fiber nonlinearities can be the dominant nonlinear effects [7,18,19,25–27].

In principle PMD impairments can be completely compensated by DSP if electronic equalizers in coherent receivers are complex enough, but the limited complexity of electronic equalizers in a real system limits its PMD tolerance [13,14]. Polarization-dependent loss (PDL) causes optical signal-to-noise-ratio (OSNR) fluctuations, re-polarizes amplified spontaneous emission (ASE) noise, and cannot be well compensated in coherent receivers. To accurately evaluate PDL impairments in a coherent optical communication system, an appropriate PDL model needs to be used [28–30]. As both PMD and PDL are statistical phenomena, their impact on optical communication systems needs to be quantified statistically, for example, using outage probabilities (OPs).

This paper reviews the recent advances in understanding fiber nonlinearities and polarization effects in coherent optical communication systems. The paper is organized as follows. In section 2, a digital coherent optical communication system is briefly described. Fiber nonlinear effects in different coherent optical communication systems are discussed in section 3. Section 4 describes the PMD tolerance of coherent optical communication systems. PDL effects in coherent optical communication systems are presented in section 5. A summary of the paper is given in section 6.

2. Digital coherent optical communication systems

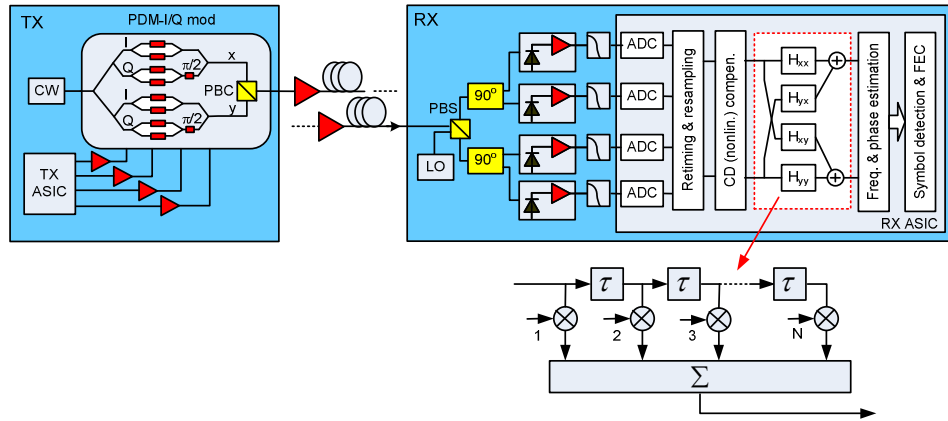


Fig. 1. Block diagram of a digital coherent optical communication system. TX: transmitter, CW: continuous wave, PDM: polarization-division multiplexed, Mod: modulator, RX: receiver, LO: local oscillator, ADC: analog-to-digital convertor, PBC/S: polarization-beam combiner/splitter, ASIC: application-specific integrated circuit, FEC: forward-error correction.

The block diagram of a digital coherent optical communication system is illustrated in Fig. 1. For simplicity, only one channel of a wavelength-division multiplexed (WDM) system is shown in the figure. Typically both polarization and phase of lightwaves are used to carry information in a coherent optical communication system to increase spectral efficiency and system capacity. In the transmitter, a continuous wave (CW) from a low linewidth laser such as an external cavity laser (ECL) is split into two parts, one for each polarization, and each part is modulated with an inphase/quadrature (I/Q) modulator by electrical signals. The electrical driving signals to the modulators are preferably generated using digital-to-analog converters (DACs) fed by an application-specific integrated circuit (ASIC), which can

generate signals with specific waveforms and spectral shapes for purposes such as to improve nonlinear tolerance and/or spectral efficiency.

During the propagation in fibers, signal polarizations are not maintained but randomly rotated. At the polarization and phase diversity receiver, the receiver signal is split by a polarization beam splitter (PBS). Each polarization of the signal after the PBS is combined with a local oscillator (LO) in a 90° hybrid. The four tributaries (x and y polarizations, I/Q branches) of combined signal and LO after the hybrids are detected by four detectors (or four pairs of balanced detectors). After anti-alias filtering, the signals are sampled and converted to digital form by ADCs. The signals are then processed by DSP, including retiming and re-sampling, CD compensation (nonlinearity compensation if allowed by the complexity of the DSP), polarization demultiplexing and PMD compensation, carrier frequency and phase estimation, symbol detection, and forward-error correction (FEC). Polarization demultiplexing and PMD compensation are performed with a butterfly equalizer, which consists of four sub-equalizers. Each sub-equalizer can be a finite impulse response (FIR) filter, as shown in Fig. 1. Apart from polarization demultiplexing and PMD compensation, the butterfly equalizer also mitigates inter-symbol interference (ISI) caused by other factors such as residual CD after the CD compensator, filtering, or nonlinearities in the system. In a real system, the DSP is typically included in an ASIC. The capability of the coherent receiver to compensate transmission impairments depends on the complexity of the DSP. For example, intra-channel nonlinearities could be significantly reduced if backward-propagation could be implemented in the receiver [21].

As CD can be completely compensated with DSP in coherent receivers, it is not necessary to have optical dispersion compensators (ODCs) in green field systems. But most existing systems have ODCs. It has been shown that fiber nonlinear effects are significantly different in coherent optical communication systems with ODCs from those in the systems without ODCs.

3. Nonlinear effects in coherent optical communication systems

Signal propagation in optical fibers can be described by the Coupled Nonlinear Schrödinger Equation (CNLSE). By averaging the nonlinear effects over the Poincaré sphere and neglecting PMD, the CNLSE can be transformed to the Manakov equation [31]

$$\frac{\partial \vec{E}}{\partial z} + \frac{i}{2} \beta_2 \frac{\partial^2 \vec{E}}{\partial t^2} - i \frac{8}{9} \gamma |\vec{E}|^2 \vec{E} = 0 \quad (1)$$

where $\vec{E} = [E_x, E_y]^T$ is the electrical field column vector, β_2 is the CD coefficient, γ is the fiber nonlinear coefficient, z is the distance along the fiber axis, t is the retarded time moving at group velocity at the carrier frequency of the signal, and $i = \sqrt{-1}$ is the imaginary unit. Without the loss of generality, we assume a WDM system with two channels, channels a and b , with the two channels having non-overlapping spectra. By neglecting four-wave mixing (FWM) between the two channels, we can obtain the equations for channel a as

$$\frac{\partial \vec{E}_a}{\partial z} + \frac{i}{2} \beta_2 \frac{\partial^2 \vec{E}_a}{\partial t^2} - i \frac{8}{9} \gamma \left(|\vec{E}_a|^2 \vec{E}_a + |\vec{E}_b|^2 \vec{E}_a + \vec{E}_b^+ \vec{E}_a \vec{E}_b \right) = 0 \quad (2)$$

where $\vec{E}_a$ and $\vec{E}_b$ are electrical field vectors of channels a and b , $\vec{E}^+ = [E_x^*, E_y^*]$ is the transpose conjugate of $\vec{E}$. In the parenthesis, the first term is self-phase modulation (SPM), including intra-channel FWM (IFWM) and intra-channel cross-phase modulation (IXPM). The second term is polarization independent cross-phase modulation (XPM), and the third term is polarization dependent XPM. Note that while SPM does not depend on the polarization, XPM is polarization dependent. The third nonlinear term is the same as the second nonlinear term when the two channels have the same polarization and it is zero when they are orthogonally polarized. For single-polarization channels, this means that XPM

between two channels with parallel polarizations is twice that with orthogonal polarizations. The last two terms in Eq. (2) show that XPM between channels also causes cross-polarization modulation (XPolM). A convenient way to describe XPolM is to use the three-dimensional Stokes vectors in the Stokes space [32],

$$\frac{d\vec{S}_a}{dz} = \frac{8}{9} \gamma (\vec{S}_a \times \vec{S}_b), \quad (3)$$

where $\vec{S}_a$ and $\vec{S}_b$ are the Stokes vectors of channel a and channel b , respectively. Equation (3) shows that the nonlinear interaction between channels modifies the state of polarization (SOP) of each channel and causes the Stokes vector of each channel to precess around each other. When channels are loaded with signals of amplitude, phase or polarization modulation, and fiber CD is present, the amplitude and SOP of each channel generally changes with time, and XPolM will generate time dependent nonlinear polarization scattering. Nonlinear polarization scattering causes the SOP to change at the speed of the symbol rate, which is hard to follow with DSP in coherent receivers and may induce severe crosstalk between two polarization tributaries.

In most coherent optical communication systems without inline ODCs, as signals are rapidly spread in time due to large accumulated CD, the dominant nonlinear effects are intra-channel nonlinearities such as IFWM and IXPM. When coherent optical communication systems have inline ODCs such as dispersion compensation fiber (DCF), the dominant nonlinear effects are inter-channel nonlinearities, including inter-channel XPolM and inter-channel XPM, but whether XPolM or XPM is dominant depends on the actual system parameters.

3.1 System model

Using numerical simulations, we investigate fiber nonlinearities in three coherent optical communication systems, a homogeneous polarization-division-multiplexed (PDM) quadrature-phase-shift-keyed (QPSK) system, a hybrid PDM-QPSK and OOK system, and a homogeneous PDM 16-ary quadrature-amplitude modulation (PDM-16QAM). The transmission system model is shown in Fig. 2. The system has seven channels with a channel spacing of 50 GHz. The transmitters can generate either 28-Gbaud QPSK, 28-Gbaud 16QAM, or 10-Gb/s OOK signals, depending on the system we study. The transmission line consists of 10 spans of standard single mode fiber (SSMF) with a CD coefficient of 17.0 ps/(nm.km), a nonlinear coefficient of 1.17 (km.W)^{-1} and a loss coefficient of 0.21 dB/km. The span length is 100 km. It has been shown that more than twenty channels are needed to accurately assess the performance of a coherent system with fibers of low CD and at high nonlinear penalties [33,34]. We think seven channels are sufficient for the studies in this paper due to two reasons: 1) we use SSMF spans, which have high CD, and 2) we focus on nonlinear effects in coherent systems with and without DCF, not absolute performance such as achievable Q^2 factor or transmission distances. In the homogenous PDM-QPSK and in the hybrid PDM-QPSK and OOK systems, an EDFA after each span is used to compensate for the transmission loss, while in the homogeneous PDM-16QAM system transmission loss is compensated for by hybrid Raman/EDFA to improve the delivered OSNR. Two different dispersion maps are studied and compared, one with legacy dispersion management supporting the 10-Gb/s OOK channels, and the other one without ODCs. In the dispersion-managed (DM) system, the CD in each span is compensated by DCF with a residual dispersion per span (RDPS) of 30 ps/nm and dispersion pre-compensation is optimized for coherent channels, which is about -400 ps/nm. The net residual dispersion after transmission is compensated in the electrical domain by DSP in the coherent receiver. The dispersion map for the DM system used here is a typical map for a direct-detection optical communication system, and no effort is made to optimize the dispersion map, reflecting upgrade scenarios of legacy 10-Gb/s systems to higher rates. In the system without any ODCs, the CD is entirely compensated in the electrical domain in the

coherent receivers. The dotted line modules in Fig. 2 are used in some of the discussed system configurations when explicitly noted.

Note that in the simulations, SOPs of all the channels are set in the same direction unless explicitly stated. It has been shown that system performance varies with the SOP changes of channels, but for systems with SSMF and penalties less than 2 dB, the performance variation is small [34].

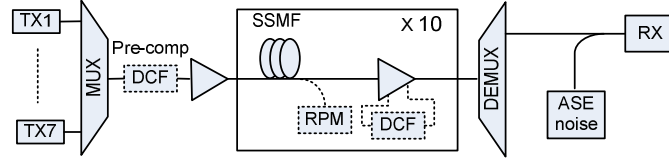


Fig. 2. Transmission system model. TX: transmitter, RX: receiver, MUX: multiplexer, DEMUX: demultiplexer, RPM: Raman pump module. The modules in dotted lines are used for some of the studied configurations, as indicated in the text.

In the simulations, no inline noise is added and ASE is loaded at the receiver. This is justified by previous studies, which showed that nonlinear phase noise induced by nonlinear signal-noise interactions is negligible in such systems [6,7]. We first propagate 1024 symbols in the transmission line, which is sufficient to capture the nonlinear interaction for the systems studied here [35], then ASE noise loading at the receiver is used to calculate a bit error rate (BER). We assume no phase noise, neither for the transmit laser nor for the LO, and no nonlinearity compensation is considered here.

3.2 Homogeneous PDM-QPSK system

For the PDM-QPSK coherent receiver used here, the butterfly equalizer has 13 taps and is optimized with the constant modulus algorithm (CMA). Carrier phase recovery is performed using the Viterbi-Viterbi phase estimation method with block length of 10, and the BER is calculated with differential decoding. In a homogeneous PDM-QPSK system with DCF, where all the channels carry PDM-QPSK signals, the amplitudes for all the channels are almost constant (neglect transient between symbols) and the same after each span and the dominant nonlinear effect is XPolM induced nonlinear polarization scattering. Figure 3 shows the required OSNR at a BER of 10^{-3} after 1000-km transmission versus launch power per channel for both 42.8-Gb/s and 112-Gb/s non-return-to-zero (NRZ) PDM-QPSK coherent systems with and without inline DCF. To separate the penalty caused by XPolM from that by XPM, we also plot the results of one 42.8-Gb/s and 112-Gb/s NRZ-PDM-QPSK channel surrounded by six 21.4-Gb/s and 56-Gb/s NRZ single-polarization (SP) QPSK channels, which have the same symbol rate as the PDM-QPSK channel. As shown in Fig. 4, the SOPs of NRZ-PDM-QPSK in different symbols are at S_2 , S_3 , $-S_2$ and $-S_3$ on the Poincaré sphere (in x and y polarizations in the Jones space). We set the SOPs of the six surrounding NRZ-SP-QPSK channels at S_1 (in x polarization in the Jones space). By doing this, at the same launch power, the average XPM on the center NRZ-PDM-QPSK channel from the surrounding NRZ-SP-QPSK and NRZ-PDM-QPSK channels are the same. As NRZ-SP-QPSK has constant amplitude (not considering the transient between symbols) and its SOP is the same in different symbols, XPolM from the surrounding NRZ-SP-QPSK channels cause little nonlinear polarization scattering in the middle PDM-QPSK channel. Therefore there is almost no XPolM induced penalty for the PDM-QPSK channel when it is surrounded by SP-QPSK channels in Fig. 3.

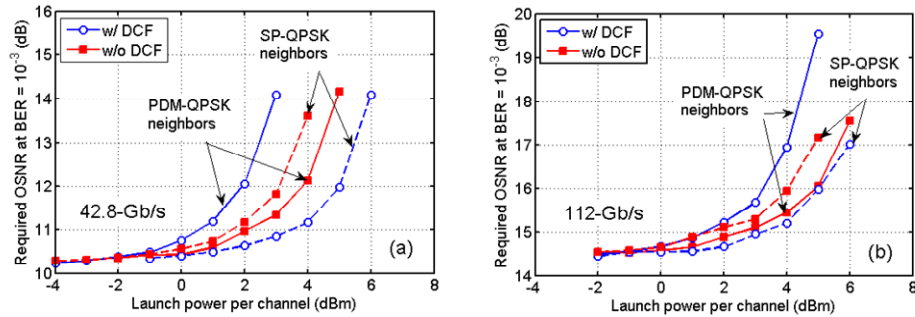


Fig. 3. Required OSNR at a BER of 10^{-3} after 1000-km transmission versus launch power per channel for the 42.8-Gb/s (a) and 112-Gb/s NRZ-PDM-QPSK (b) coherent systems with and without inline DCF. Solid lines are for the results when all the channels are PDM-QPSK channels and dotted lines are the results when the center PDM-QPSK channel is surrounded by six SP-QPSK channels.

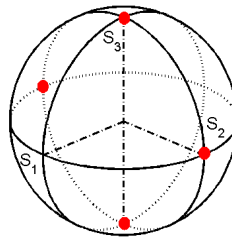


Fig. 4. SOP of PDM-QPSK on the Poincaré sphere. The symbols on x and y polarizations are aligned.

Figure 3 shows that when all the channels carry NRZ-PDM-QPSK signals, the system with inline DCF performs worse than that without DCF. For 42.8-Gb/s and 112-Gb/s NRZ-PDM-QPSK, the maximum launch powers at 1-dB OSNR penalty for the systems with DCF are about 2-dB and 1.5-dB lower than those without DCF, respectively. However, when the surrounding channels carry NRZ-SP-QPSK signals, the systems with inline DCF perform better than those without DCF. At 1-dB OSNR penalty, the maximum launch powers for the systems with DCF are about 2.0-dB and 1.0-dB higher than those without DCF for the 42.8-Gb/s and 112-Gb/s NRZ-PDM-QPSK, respectively. In addition, Fig. 3 also shows that when the surrounding channels are changed from NRZ-SP-QPSK to NRZ-PDM-QPSK, the allowed launch power is reduced by about 3 dB and 2 dB for the 42.8-Gb/s and 112-Gb/s systems with DCF, respectively, whereas it is increased by about 1 dB for both 42.8-Gb/s and 112-Gb/s systems without DCF. Those indicate that XPolM is the dominant nonlinear effect in the homogeneous NRZ-PDM-QPSK system with DCF and it is XPolM that makes homogeneous PDM-QPSK systems with DCF perform worse than those without DCF. The reason why the PDM-QPSK channels cause less inter-channel penalty than SP-QPSK channels in the systems without DCF is because that the impact of the inter-channel XPM is much larger than XPolM in the systems without DCF and the peak powers of PDM signals are smaller than those of SP signals for a given average power due to different data in the two polarizations for PDM signals. Note that the performance difference between the 112-Gb/s PDM-QPSK systems with DCF and without DCF is smaller than that between the 42.8-Gb/s systems because of the increased symbol rate.

The above conclusion is further confirmed in Fig. 5, which shows a reduction in the degree of polarization (DOP) of 21.4-Gb/s and 56-Gb/s SP-QPSK reference channels caused by XPolM induced signal depolarization from six surrounding 42.8-Gb/s and 112-Gb/s PDM-QPSK channels. For the systems with inline DCF, the DOP decreases more rapidly with the launch power than for those without DCF, indicating that the nonlinear polarization scattering

is much larger in systems with DCF than that without DCF. We also note that the depolarization in the 112-Gb/s PDM-QPSK systems is smaller than that in the 42.8-Gb/s systems due to the increase of the symbol rate.

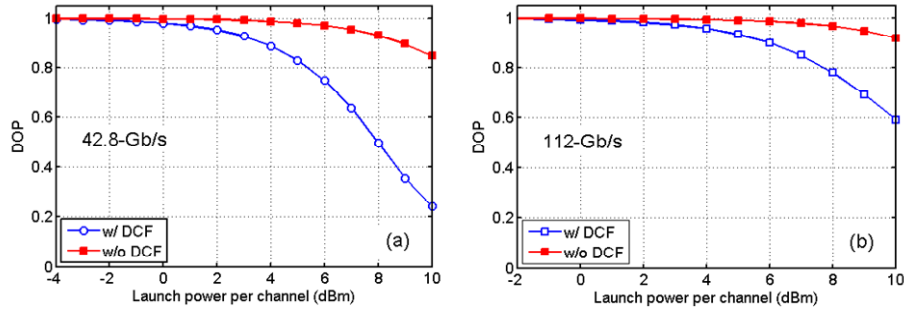


Fig. 5. XPoIM induced depolarization in the 42.8-Gb/s (a) and 112-Gb/s (b) PDM-QPSK systems with and without inline DCF after 1000-km transmission. DOP: degree of polarization.

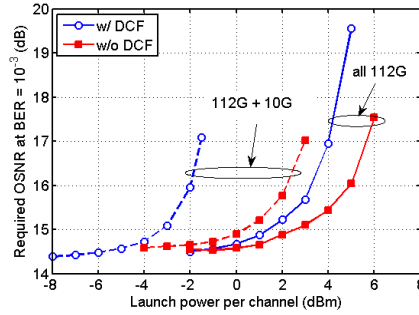


Fig. 6. Required OSNR at a BER of 10^{-3} after 1000-km transmission versus launch power per channel for 112-Gb/s NRZ-PDM-QPSK co-propagating with six 10-Gb/s OOK channels.

3.3 Hybrid PDM-QPSK and 10-Gb/s OOK system

It has been shown that co-propagating 10-Gb/s OOK-channels can severely degrade the performance of PDM-QPSK channels in hybrid PDM-QPSK and 10-Gb/s OOK systems [25,26,36]. In such systems, the penalty is mainly caused by inter-channel XPM, not XPoIM, because of the non-constant amplitude of 10-Gb/s OOK signals. Figure 6 gives the performance of 112-Gb/s PDM-QPSK in a hybrid system, where one 112-Gb/s NRZ-PDM-QPSK channel is surrounded by six 10-Gb/s NRZ-OOK channels at 50-GHz channel spacing. The SOPs of the 10-Gb/s OOK channels are set at S_1 in the Stokes space (x polarization in the Jones space) to maximum XPoIM effects [25,32], and the SOP of the PDM-QPSK channel is the same as that shown in Fig. 4. As shown by Eq. (3), the XPoIM between two channels is the largest when their SOPs are perpendicular to each other in the Stokes space. For comparison, we also plot the performance of 112-Gb/s PDM-QPSK in a homogeneous PDM-QPSK system. Due to large inter-channel nonlinearities from the OOK channels, the maximum launch powers for the systems with DCF and without DCF at 1-dB penalty are reduced by 5 dB and 3 dB, respectively.

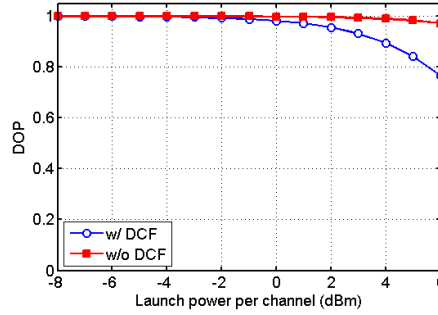


Fig. 7. XPolM from 10-Gb/s OOK channels induced depolarization in the 112-Gb/s PDM-QPSK channel with and without inline DCF after 1000-km transmission. DOP: degree of polarization.

In Fig. 7, we depict the DOP reduction of a 56-Gb/s SP-QPSK reference channel caused by XPolM induced signal depolarization from the surrounding six 10-Gb/s OOK channels. Comparing the result with that in Fig. 5(b) (note the different scale of the x-axes) shows that XPolM is larger in the hybrid system than in the homogeneous system due to the lower symbol rate (and hence the slower waveform evolution) of the 10-Gb/s OOK channels. The fact that XPolM is not the main degrading factor in the hybrid system can be seen from Fig. 6, where in the system with DCF at -1 -dBm per-channel launch power, the OOK channels already induce more than a 3-dB penalty on the PDM-QPSK channel, while the depolarization caused by the 10-Gb/s OOK channels at this power level (Fig. 7) is very small (DOP above 0.98), which by itself would not cause any noticeable performance degradation for the 112-Gb/s PDM-QPSK channel. Along the same lines, we see from Figs. 6 and 7 that the penalty caused by XPolM induced depolarization in the hybrid system without DCF is also small.

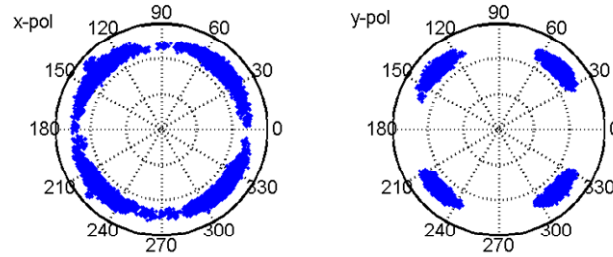


Fig. 8. Signal constellations of 112-Gb/s PDM-QPSK after co-propagating with 10-Gb/s OOK channels over 1000-km SSMF with DCF at -1 -dBm per channel launch power.

Finally the signal constellations of the 112-Gb/s NRZ-PDM-QPSK after 1000-km transmission in the hybrid system at -1 -dBm per-channel launch power are given in Fig. 8. No ASE noise and laser phase noise are added. It shows that the PDM-QPSK channel has large nonlinear phase distortions after co-propagating with the 10-Gb/s OOK channels, indicating that XPM from the neighboring 10-Gb/s OOK channels is the main nonlinear effect that causes the large performance degradation for the 112-Gb/s PDM-QPSK channel in the hybrid system. As the OOK channels are aligned with the x-polarization, XPM on the x-polarization of the PDM-QPSK signal is twice that on the y-polarization. We used a block length of 10 for the Viterbi-Viterbi phase estimation in Figs. 6 and 8. If we reduce the block length to 2, the performance of the hybrid system with DCF can be improved by 1-2 dB due to the correlation of XPM induced nonlinear phase distortions among neighboring symbols [26], which also indicates that XPM is the dominant nonlinear effect in such systems.

3.4 Homogeneous 224-Gb/s PDM-16QAM system

To get sufficient delivered OSNR for PDM-16QAM transmission, we add hybrid Raman/EDFA amplification to compensate for the transmission loss, with 15-dB on/off gain provided by the Raman amplifier. As time-interleaved return-to-zero (iRZ) has been shown to have better nonlinear tolerance than NRZ, iRZ-PDM-16QAM is used here [8,16,27]. In the PDM-16QAM receiver, polarization demultiplexing and residual distortion equalization is performed with a butterfly equalizer consisting of four 13-tap FIR filters. These are first optimized with the CMA for pre-convergence and then fine tuned with decision-directed least-mean-square (DD-LMS) algorithm. Carrier phase recovery is performed with a decision-directed phase estimation method [37]. The BER is evaluated by direct error counting with Gray coding.

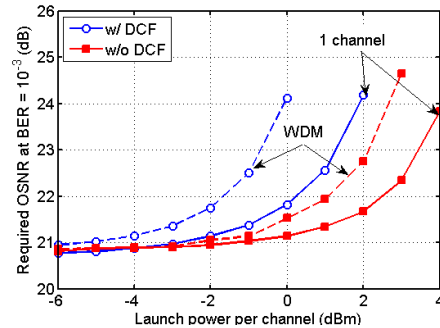


Fig. 9. Required OSNR at a BER of 10^{-3} after 1000-km transmission versus launch power per channel for single channel and WDM 224-Gb/s iRZ-PDM-16QAM systems with and without DCF.

The nonlinear transmission performance of 224-Gb/s iRZ-PDM-16QAM is given in Fig. 9 [27]. It shows that even for single-channel transmission using iRZ, the system with DCF performs worse than that without DCF and the nonlinear tolerance of single-channel transmission in the system with DCF is even smaller than that of WDM transmission in the system without DCF, which is completely different from the PDM-QPSK system. At 1-dB OSNR penalty, the maximum launch power for the single channel and WDM transmission in the system with DCF is about 2-dB and 3-dB lower than that without DCF. The poorer nonlinear transmission performance of the PDM-16QAM system with DCF is mainly caused by SPM and XPM induced nonlinear phase distortions for single-channel and WDM transmission, respectively.

Figure 10 gives the signal constellations of iRZ-PDM-16QAM after 1000-km transmission for the single-channel and WDM systems with and without DCF (only one polarization is shown and the other polarization is similar). The launch powers per channel are 3-dBm and 1-dBm for the single channel and WDM systems, respectively. As shown in Fig. 10, for the system with DCF, there are large nonlinear phase distortions in both single channel and WDM transmission, whereas for the system without DCF, there is almost no observable nonlinear phase distortion and the distortions are similar in all directions in the complex symbol plane.

The XPolM induced depolarization in the iRZ-PDM-16QAM transmission systems is depicted in Fig. 11, which is measured by the DOP of a 112-Gb/s RZ-SP-16QAM channel surrounded by six 224-Gb/s iRZ-PDM-16QAM channels. For comparison, the result of the 112-Gb/s iRZ-PDM-QPSK system with DCF is also plotted in the figure. The XPolM induced depolarization is similar in the PDM-16QAM system and in the PDM-QPSK system. The figure also indicates that the dominant nonlinear effect in the WDM PDM-16QAM system with DCF is not XPolM, but XPM, as the OSNR penalty at 1-dBm per channel launch power in the WDM system with DCF is more than 3 dB, but the XPolM induced depolarization is very small at this launch power (DOP is 9.93) and will not cause any noticeable penalties.

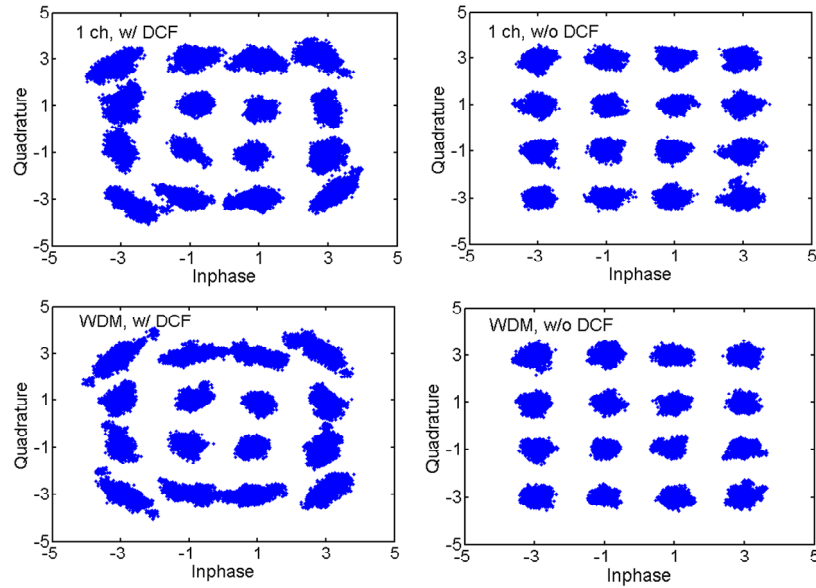


Fig. 10. Signal constellations of one polarization for single-channel and WDM 224-Gb/s iRZ-PDM-16QAM after 1000-km transmission with and without DCF and at 3-dBm and 1-dBm per-channel launch powers for single-channel and WDM transmission, respectively.

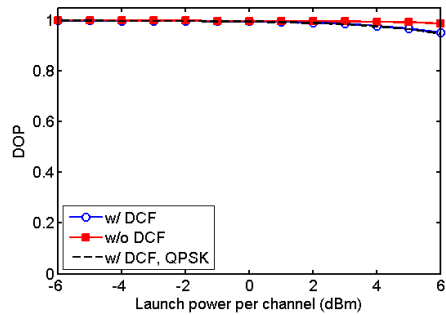


Fig. 11. XPoLM induced depolarization in the 224-Gb/s iRZ-PDM-16QAM system with and without inline DCF after 1000-km transmission, and in the 112-Gb/s iRZ-PDM-QPSK system with DCF. DOP: degree of polarization.

4. PMD effects in coherent optical communication systems

PMD was a hot research topic in the past decade and has long been considered as one of the limiting factors for optical communication systems. Various PMD compensation techniques have been proposed and demonstrated, but PMD impairments can only be partly compensated in direct-detection systems due to higher-order PMD and its stochastic characteristics. As PMD impairments in principle can be completely compensated with DSP in a coherent receiver if the DSP is complex enough, not much attention is usually being paid to PMD in coherent optical communication systems. But the limited complexity of DSP in a real system will put a limit on the amount of PMD that a coherent system can tolerate, so it is important for a system designer to know the amount of PMD that a coherent receiver with given complexity can compensate, and to understand the implications on margin allocation and system outage.

PMD can be represented by the PMD vector. Using a Taylor expansion, the PMD vector can be expressed as [38]

$$\bar{\Omega}(\omega_0 + \Delta\omega) = \bar{\Omega}_0(\omega_0) + \bar{\Omega}_\omega(\omega_0)\Delta\omega + \frac{1}{2}\bar{\Omega}_{\omega\omega}(\omega_0)\Delta\omega^2 + \dots \quad (4)$$

where $\bar{\Omega}_0 = \Delta\tau\hat{p}$ is the 1st-order PMD, $\Delta\tau$ is the differential group delay (DGD), $\hat{p}$ is the principal state of polarization (PSP), $\bar{\Omega}_\omega$ is the 2nd-order PMD, and the remaining terms are higher-order PMD. The 2nd-order PMD has two terms, $\bar{\Omega}_\omega = \Delta\tau_\omega\hat{p} + \Delta\tau\hat{p}_\omega$, where the 1st-term is the polarization dependent chromatic dispersion (PCD) and the 2nd term is the depolarization. In an optical communication system, PMD causes pulse broadening and signal depolarization. As signal depolarization introduces crosstalk between two polarization tributaries, it will cause much larger penalties than the pulse broadening effect for a PDM signal.

Various PMD emulators (PMDEs) have been used to evaluate PMD effects in optical communication systems, including 1st-order PMDEs, 2nd-order PMDEs and all-order PMDEs. It has been shown that using different PMDEs to evaluate PMD penalties in an optical communication system can generate significantly different results, and this is true in a coherent optical communication system as well [14,39]. Figure 12 depicts the PMD induced OPs in a 112-Gb/s NRZ-PDM-QPSK coherent system with three different PMDEs, a 1st-order PMDE, a 2nd-order PMDE that models the entire 2nd-order PMD as PCD, and an all-order PMDE, which is the concatenation of 100 waveplates. OP is widely used to quantify PMD and other polarization related impairments, and is defined as the probability that the penalty of a system exceeds an allocated margin. In Fig. 12, the butterfly equalizer has 7-tap FIR filters, and 0.5-dB OSNR margin at a BER of 10^{-3} is used for the OP simulation [14]. The figure shows that using a 1st-order PMDE underestimates PMD impairments (and the PMD tolerance will hence be overestimated by 10-20%), while using the 2nd-order PMDE that models 2nd-order PMD entirely as PCD significantly overestimates PMD impairments.

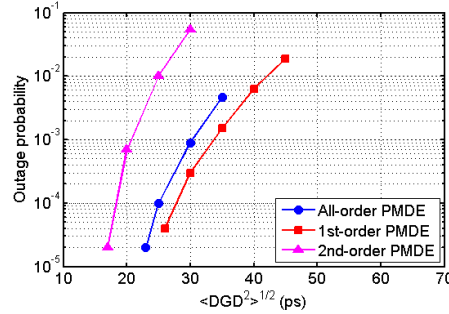


Fig. 12. PMD induced OPs for a 112-Gb/s NRZ-PDM-QPSK system with a 7-tap butterfly equalizer for different PMDEs. A 0.5-dB OSNR margin is used.

PMD tolerance can be increased by increasing the length of the butterfly equalizer, as shown in Fig. 13, which depicts the PMD induced OPs in a 112-Gb/s NRZ-PDM-QPSK coherent system with butterfly equalizers of various lengths. As expected, doubling the tap number almost doubles the PMD tolerance. At an OP of 10^{-5} , the tolerable PMD (average DGD) is about 22 ps, 39 ps, and 44 ps for equalizers with 7, 12, and 14 taps, respectively.

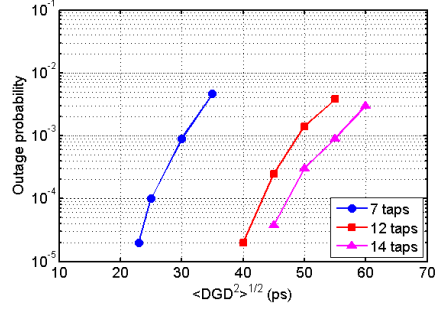


Fig. 13. PMD induced OPs for a 112-Gb/s NRZ-PDM-QPSK system with the butterfly equalizer of different tap numbers. A 0.5-dB OSNR margin and all-order PMDE are used.

5. PDL effects in coherent optical communication systems

Although CD and PMD can be easily compensated in a coherent receiver by powerful DSP, PDL effects cannot be well compensated in a coherent receiver due to non-unitary nature of PDL [28–30]. PDL causes signal power and OSNR fluctuations and re-polarizes ASE noise. In addition, it induces loss of orthogonality and power/OSNR imbalance between the two polarizations of a PDM signal. If the polarization of a PDM signal is not aligned with the PDL axis at the input, the two originally orthogonal polarizations will become non-orthogonal at the output [40]. Although the non-orthogonality between the two polarizations can be corrected by electronic equalizers, some penalties will be introduced. When the polarization of a PDM signal is aligned with the PDL axis at the input, one polarization is improved, but the other polarization is degraded and the overall performance is mainly determined by the degraded polarization tributary.

There are two PDL models to study PDL effects in a coherent system. One is a lumped model and the other is a distributed model, as shown in Fig. 14. In the lumped model, there is one PDL emulator (PDLE) and ASE noise is loaded at the receiver. In the distributed model, many PDLEs and ASE noise sources are distributed along a link, with random polarization rotations between the PDLEs. The lumped model, usually used in lab tests, is simple and helpful to understand some PDL effects but it does not include all PDL effects such as repolarization of ASE noise. Typically a polarization controller (PC) or polarization scrambler is inserted before the PDLE in the lumped model to get the PDL penalty at a particular input SOP or an average PDL penalty. The distributed model is similar to a real system and automatically takes into account all PDL effects, but it needs more resources to build a distributed PDLE and the distributed PDL model has to be analyzed statistically.

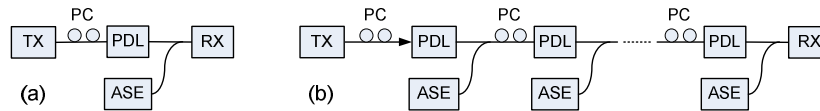


Fig. 14. PDL models. (a) lumped model, (b) distributed model. PC: polarization controller.

In the lumped model, the ASE noise power is fixed at the receiver, while in a real system, ASE noise is generated along the link and when PDL attenuates the signal, it attenuates the noise as well. Therefore, the lumped model over-estimates OSNR variations and thus PDL penalties [30]. This is shown in Fig. 15, which depicts the probability distribution function of OSNR variations in one polarization calculated with the lumped model and distributed model at a root mean square (RMS) PDL value of 3 dB. For the distributed model, 20 PDL elements are used, and PDL and ASE noise are equally distributed along the link. The figure shows that the lumped model generates much larger OSNR variations than the distributed model.

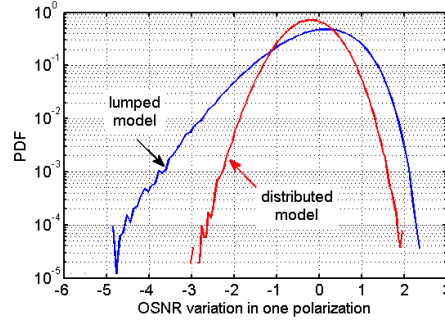


Fig. 15. Simulated probability density function (PDF) of PDL induced OSNR variations in one polarization using the lumped model and the distributed model at an RMS PDL value of 3 dB.

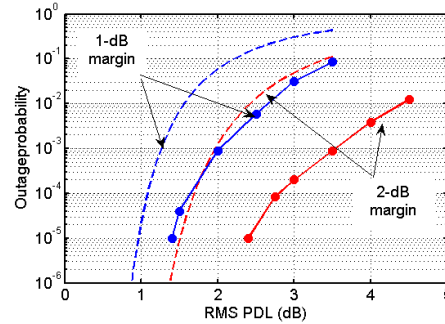


Fig. 16. PDL induced OPs at $\text{BER} = 10^{-3}$ versus RMS PDL in a 112-Gb/s PDM-QPSK system using the distributed model (symbols and solid lines) and lumped model (dashed lines).

As PDL is a statistical phenomenon like PMD, its impact on an optical communication system also needs to be quantified statistically. PDL induced OPs at $\text{BER} = 10^{-3}$ in a 112-Gb/s NRZ-PDM-QPSK system with 1- and 2-dB OSNR margins using the distributed model are given in Fig. 16. For comparison, the results using the lumped model are also given [30]. The figure clearly shows that the lumped model significantly over-estimates PDL penalties. At an OP of 10^{-5} , the tolerable RMS PDL with 1- and 2-dB OSNR margins obtained using the distributed model is about 1.4 and 2.4 dB, respectively, while it is about 1.0 and 1.5 dB, respectively, if the lumped model is used.

6. Summary

We have reviewed recent advances in understanding fiber nonlinearity and polarization effects in coherent optical communication systems. While the dominant nonlinear effects in coherent optical communication systems without ODCs are mostly intra-channel nonlinearities, the dominant nonlinear effects are inter-channel nonlinearities in coherent optical systems with inline DCF. In coherent optical communication systems with DCF, when modulation formats of constant amplitude such as QPSK are used, the dominant nonlinear effect is XPolM, which generates nonlinear polarization scattering and induces severe crosstalk between polarization tributaries, whereas when the channels carry non-constant amplitude modulation formats such as hybrid QPSK and OOK or 16QAM, XPM is the dominant nonlinear effect. We quantified the PMD tolerance of coherent optical communication systems using OPs. We showed that due to the limited complexity of DSP in the coherent receiver, the PMD tolerance of a coherent receiver is limited. PDL effects cannot be well compensated in a coherent receiver due to the non-unitary nature of PDL. We described two PDL models and showed that the lumped PDL model significantly over-estimates PDL penalties, and thus in order to accurately

evaluate the PDL penalties in coherent optical communication systems, the distributed PDL model has to be used.

Acknowledgments

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